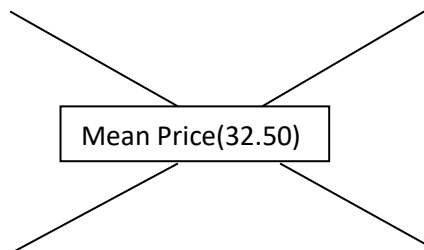


Assignment_Alligation or Mixture

Eg.1. In what proportion must a shopkeeper mix two varieties of rice costing Rs.31 per kg and Rs.36 per kg. So that mixture be worth Rs.32.50 a kg?

Solution: C.P. of 1kg of cheaper rice (Rs.31)

C.P. of 1 kg of dearer rice (Rs. 36)



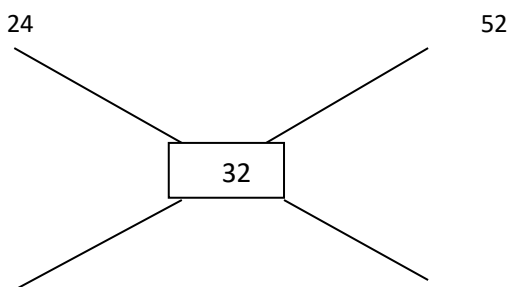
$$(d - m) = 36 - 32.50 = 3.50$$

$$(m - c) = 31 - 32.50 = 1.50$$

By Alligation rule: $\frac{\text{Quantity of Cheaper}}{\text{Quantity of Dearer}} = \frac{\text{C.P of Dearer} - \text{Mean Price}}{\text{Mean Price} - \text{C.P of Cheaper}} = \frac{3.50}{1.50}$ Required ratio = 7 : 3

Eg.2. How many litre of petrol at Rs.52 per litre must a vendor mix with 25 litres of kerosene at Rs.24 per litre so that he may, on selling the mixture at Rs.40 per litre gain 25% on the outlay?

Solution: Cost price of the mixture = $40 \times \frac{100}{125} = \text{Rs.32 per litre}$



$$(d - m) = 52 - 32 = 20$$

$$(m - c) = 32 - 24 = 8$$

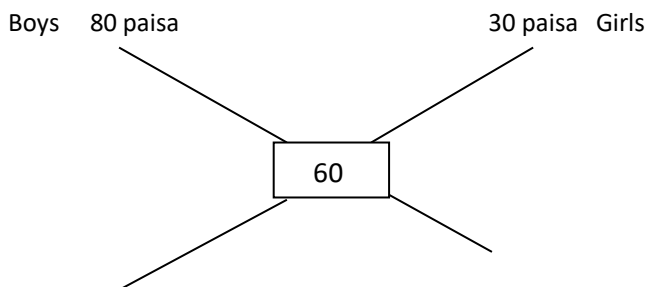
$$\text{Ratio} = 20:8 = 5:2$$

Thus for every 5 litre of petrol at Rs.52 he must mix 2litres of petrol at Rs.24 per litre

So, for 25 litres of petrol at Rs.24 that is to be mixed with petrol at Rs.52 = $25 \times \frac{2}{5} = 10$ litres.

Eg.3. There are 130 children in a colony. Rs. 78/- are distributed among them so that each boy gets 80 paisa and each girl gets 30 paisa. Find the number of boys and girls in that colony.

Solution: Here the Alligation is applicable for "Money per boy or girl" , Mean value per child = $\frac{7800}{130} = 60 \text{ paisa}$



Assignment _Alligation or Mixture

Boys : girls = 3 : 2 $\therefore$ Number of boys = $\frac{130}{3+2} \times 3 = 78$ $\therefore$ Number of girls = $130 - 78 = 52$

Section 1

1. In what ratio must a shopkeeper mix two varieties of rice at Rs. 68 per kg with another at Rs. 75 per kg to get a mixture worth Rs. 72 per kg? Ans- 3 : 4
2. In what ratio must a shopkeeper mix two varieties of sugar at Rs. 135 per kg with another at Rs. 149 per kg to get a mixture worth Rs. 145 per kg? Ans- 2 : 5
3. In what ratio must water be mixed with Milk costing Rs.54 per litre to obtain a mixture worth of Rs. 48 per litre? Ans- 1:8
4. In what ratio must Milk costing Rs.36 per litre be mixed with water to obtain a mixture worth of Rs. 20 per litre? Ans- 5:4
5. In what ratio must water be mixed with milk to gain 10% on selling the mixture at cost price? Ans- 1:10
6. In what ratio must water be mixed with milk to gain 33 $\frac{1}{3}$ % on selling the mixture at cost price? Ans- 1:3
7. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 12.5%. The percentage of water in the mixture is: Ans- 11 $\frac{1}{9}$ %
8. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 13 $\frac{1}{3}$ %. The percentage of water in the mixture is: Ans-11 $\frac{13}{17}$ %
9. A person purchased milk for Rs.62 per litre. In what ratio should water be mixed in the milk so that after selling the mixture at Rs.55.20/litre he may get a profit of 15 %? Ans- 7 : 24
10. A person purchased spirit for Rs.25 per litre. In what ratio should water be mixed in the spirit so that after selling the mixture at Rs.21/litre he may get a profit of 5 %? Ans- 1:4
11. Cost price of milk adulterated with 12 litres of water is Rs 42 per litre. If the cost price of pure milk is Rs 56 per litre, find the quantity of milk in the mixture? 36 litres.
12. A shopkeeper sells wine adulterated with 15 litres of water. Cost price of the adulterated wine for the shopkeeper is Rs 63 per litre. If he purchased pure wine for Rs 72 per litre, find the quantity of wine in the mixture? 105 litres.
13. Cost price of spirit adulterated with 20 litres of water is Rs 80 per litre. If the cost price of pure spirit is Rs 90 per litre, find the quantity of spirit in the mixture? 160 litres.
14. Gold is 19 times as heavy as water and copper 9 times. In what ratio should these metals be mixed so that the mixture may be 15 times as heavy as water? Ans- 3:2
15. How much papaya seeds at Rs. 12 a kg. should be added to 48 kg. of pepper at Rs. 78 a kg. so that the mixture be worth Rs. 60 a kg.? Ans- 18 kg
16. How much groundnut powder at Rs. 90 a kg. should be added to 56kg. of cashew powder at Rs. 310 a kg. so that by selling the mixture for Rs. 280 a kg, a person will gain 12% profit? Ans- 21 kg
17. One quality of turmeric at Rs. 86 per kg is mixed with another quality of turmeric in the ratio of 4:7. If the mixture so formed be worth Rs.72 per kg, what is the rate per kg of the second quality of turmeric? Rs. 64/-
18. One quality of cumin at Rs. 212 per kg is mixed with another quality of cumin in the ratio of 3:5. If the mixture so formed be worth Rs.202 per kg, what is the rate per kg of the second quality of cumin? Rs. 196/-
19. A sum of Rs. 30.50 is made up of 100 coins which are either 50-paise or 20-paise coins. How many are of 50 P.? Ans- 35
20. A sum of Rs. 38 is made up of 70 coins which are either Rs 1 or 20-paise coins. How many are of 20 P.? Ans- 40
21. In a poultry farm, there are a certain number of hens and goats. If heads are counted, there are 80 and if legs are counted, there are 200. How many goats are there? Ans- 20

Assignment _Alligation or Mixture

22. In a poultry farm, there are a certain number of hens and goats. If heads are counted, there are 125 and if legs are counted, there are 430. Find the difference between the number of goats and hens? Ans- 55
23. A man has 30,000 rupees to lend on loan. He lent some of his capital to Mohan at an interest of 20% per annum and the rest to Suresh at an interest rate of 12% per annum. At the end of one year he got 17% of his capital as interest. How much did he lend to Mohan ? Ans- Rs. 18,750/- (Ratio -5:3)
24. Some amount out of Rs. 21000 was lent at 8% p.a. and the remaining at 4% p.a. If the total simple interest from both the fractions in 5 years was Rs. 6300, The sum lent at 4% p. a. was _____. Ans- Rs. 10500
25. A goldsmith has two qualities of gold one of 16 carats and another of 20 carats. In what proportion should he mix both to make an ornament of 17 carats. Ans: 3 : 1
26. A sum of Rs 41 was divided among 50 boys and girls. Each boy gets 90 paise and each girl 65 paise. The number of boys is - _____. Ans- 34
27. In a mixture of 60 litres, the ratio of milk and water is 2:1. If the ratio of milk and water is to be 1:2, then the amount of water to be further added is _____. Ans- 60 litres
28. A vessel contains 56 litres of a mixture of milk and water in the ratio 5:2. How much water should be mixed with it so that the ratio of milk to water be 4:5? Ans: 34 litres
29. There are 170 children in a colony. Rs. 136/- are distributed among them so that each boy gets 85 paisa and each girl gets 75 paisa. Find the number of boys and girls in that colony. A:85 & 85
30. The average age of all the officers in an organisation is 48. If the average age of 300 senior officers is 52 and the average age of all the other junior officer is 46, what is the total number of officers in the organisation? A: 900
31. The average salary of all the employees in an organisation is Rs 25000. If the average salary of 400 female employees is 20500 and the average salary of all the male employees is Rs 27000, what is the total number of employees in the organisation? A: 1300
32. A man purchased a CD and a book for Rs 2500 and he sold both the articles at a profit of 15%. He earned 20% profit on the CD and 7.5% profit on the book; find the cost of the book? A: Rs 1000
33. A man purchased a bike and a scooter for Rs 45,000. He sold both the vehicles and earned a profit of 12%. He earned 8% profit by selling the bike and 17% profit by selling the scooter; find the difference between the cost of bike and scooter?
A: Rs 5000 (25000 and 20000)
34. A man purchased a Computer and a printer for Rs 42000. He sold both the articles and earned a profit of 17%. He earned 37% profit and 9% profit by selling the computer and the printer respectively. Find the ratio between the cost prices of the computer and the printer. A: 2:5
35. Can A contain milk and water in the ratio of 1:3 and Can B contains milk and water in the ratio of 1:1. What will be the ratio in which these mixtures should be mixed to get a new mixture of milk and water in the ratio of 3:5? A: 1:1

Assignment_Alligation or Mixture

Section 2

I. Directions (Q.1-5): Study the following information carefully to answer the questions that follow:

All the three shopkeepers viz. A, B and C mixes wheat flour with gram flour. A, B and C purchases gram flour at the rate of Rs. 40 per kg, Rs. 50 per kg and Rs. 60 per kg respectively. All of them purchase wheat flour from the same wholesaler at the rate of Rs. 10 per kg.

A sells the mixture for Rs. 42 per kg and earns a profit of 20%. B sells the mixture for Rs. 41.40 per kg at a loss of 10%. C mixes gram flour and wheat flour in the ratio of 4:1 and earns a profit of 8% by selling it.

1. From which shopkeeper, you will get the gram flour at the cheapest rate. (Assume that gram flour means mixture of gram flour and wheat flour mixed by the shopkeeper).

- 1) A 2) B 3) C 4) cannot be determined 5) None of these

2. Shopkeeper A mixes the gram flour and wheat flour in the ratio of

- 1) 2:1 2) 3:1 3) 4:1 4) 5:1 5) None of these

3. Shopkeeper B mixes the wheat flour and gram flour in the ratio of

- 1) 9:1 2) 3:1 3) 1:9 4) 1:4 5) None of these

4. Cost price of the mixture of gram flour and wheat flour is minimum for which of the shopkeeper.

- 1) A 2) B 3) C

4) cannot be determined 5) None of these

5. If a new shopkeeper in that area purchases the gram flour from shopkeeper B and shopkeeper C, in what ratio he should mix the two mixtures so that he get a mixture of gram flour and wheat flour of ratio 6:1.

- 1) 3:4 2) 4:3 3) 3:2 4) 5:2 5) None of these

II. Directions (Q.1-5): Study the following information carefully to answer the questions that follow:

Four milkmen A, B, C and D purchases pure milk at Rs. 36 per litre and mixes water with the milk. A mixes 2 litre water in 8 litres of milk. B mixes some water with pure milk and earns a profit of 10% by selling the mixture at Rs. 33 per litre. C sells the mixture at the cost price for milk and earns a profit of 20%. D mixes water with the milk in such a way that cost price of the mixture for him is Rs. 27.

1. In what ratio milkmen B mixes water with the pure milk.

- 1) 2:1 2) 3:1 3) 4:1 4) 1:5 5) None of these

2. If you want to purchase 3 litres of milk from D, how much you have to pay?

- 1) Rs. 99 2) Rs. 108 3) Rs. 123

- 4) data inadequate 5) None of these

3. If A sells 5 litres of mixtures for Rs. 150, what is the profit earned by him?

- 1) Rs. 8 2) Rs. 12 3) Rs. 6

- 4) Rs. 14 5) None of these

4. In what ratio milkmen D mixes water with the pure milk.

- 1) 2:1 2) 3:1 3) 4:1 4) 1:5 5) None of these

5. If A wants to earn 20% profit by selling the mixture, what should be the selling price per kg?

- 1) Rs. 34.56 2) Rs. 36 3) Rs. 38.20

- 4) Rs. 35.50 5) None of these

Section 2 Answer

I.

	A		B		C	
	G flour	W flour	G flour	W flour	G flour	W flour
CP per kg (Rs.)	40	10	50	10	60	10
ratio	5	1	9	1	4	1

	A	B	C
CP of mixture a kg (Rs.)	35	46	50
SP of mixture a kg (Rs.)	42	41.40	54

1. 2 2. 4 3. 3 4. 1 5. 2

II.

	A		B		C		D	
	Milk	Water	Milk	Water	Milk	Water	Milk	Water
CP per kg (Rs.)	36	0	36	0	36	0	36	0
ratio	4	1	5	1	5	1	3	1

	A	B	C	D
CP of mixture a kg (Rs.)	28.8	30	30	27
SP of mixture a kg (Rs.)	-	33	36	-



Assignment_Alligation or Mixture

1. 4 2. 4 3. 3 4. 5 ; 1:3 5. 1